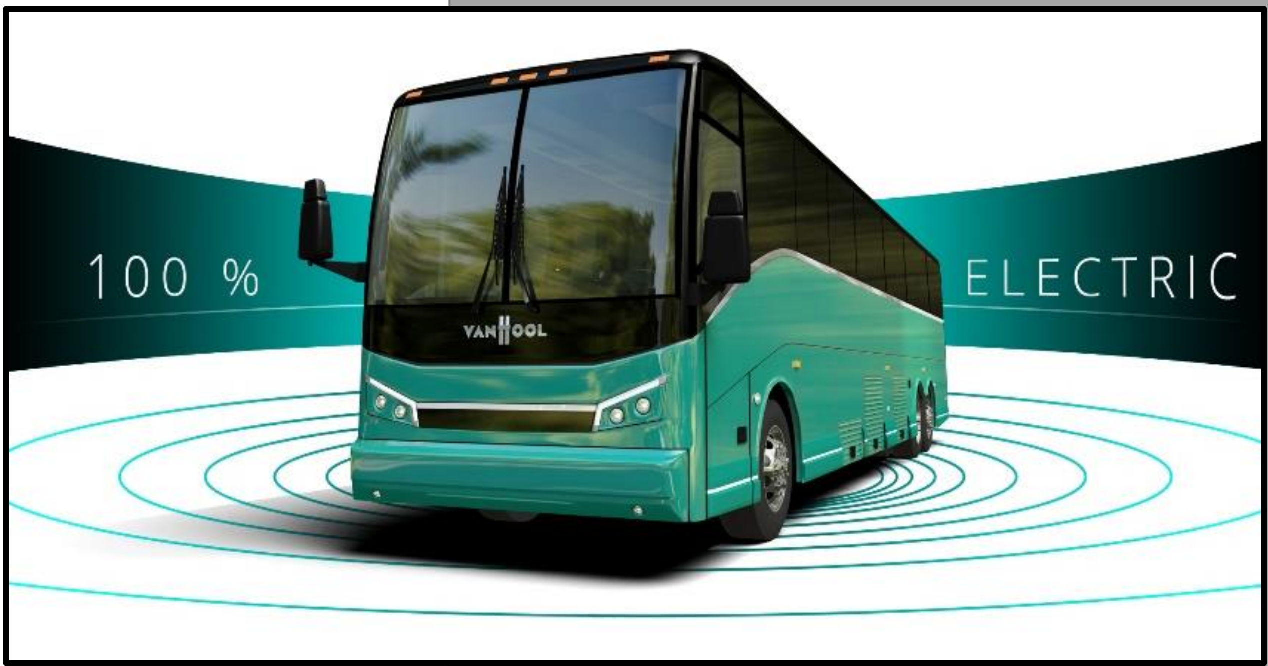
Lithium-Ion (Li-ion) batteries

CX45E (FULL ELECTRIC)

OVER-THE-ROAD BUS

EMERGENCY RESPONSE GUIDE (ERG)

INFORMATION FOR FIRST AND
SECOND RESPONDERS



Document Identification Number: ERG00036
Second edition
Date: July 14th, 2021

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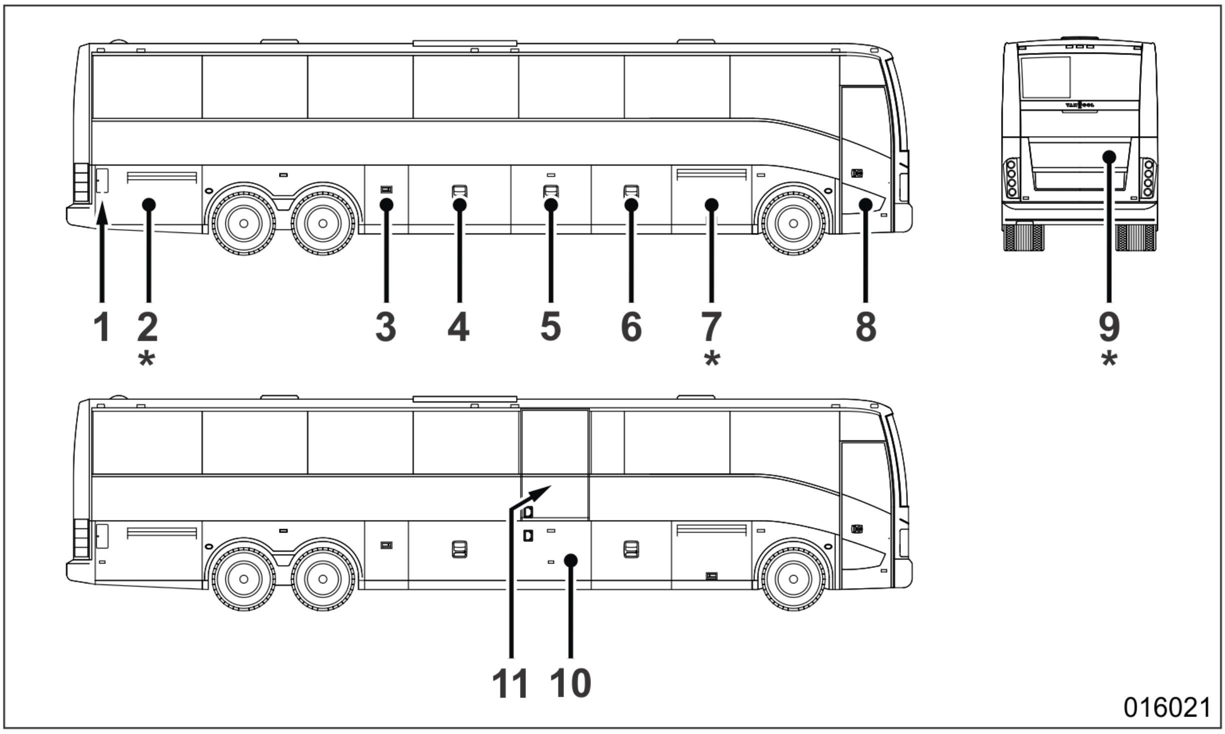
Right-hand side and rear view

WARNING!

Compartments with an asterisk (*) contain high-voltage components.



1. Charging socket for traction batteries	**7.** Power-steering fluid tank, traction batteries (one string of three)
2. Traction batteries (two strings of three)	**8.** Passenger door
3. 24V electrical system batteries, mechanical battery switch of 24V electrical system, air suspension test fittings of axle 2 and 3	**9.** Waste-water tank dump valve, toilet tank dump valve, access to junction box EK20, coolant-expansion tank of traction batteries cooling circuit, traction system emergency disconnect switch (alternative)
4. Luggage compartment, access to box with compressed-air switches and sensors, junction box EKB	**10.** Lower lift door (if present)
5. Luggage compartment	**11.** Upper lift door (if present)
6. Luggage compartment, junction box EKV	

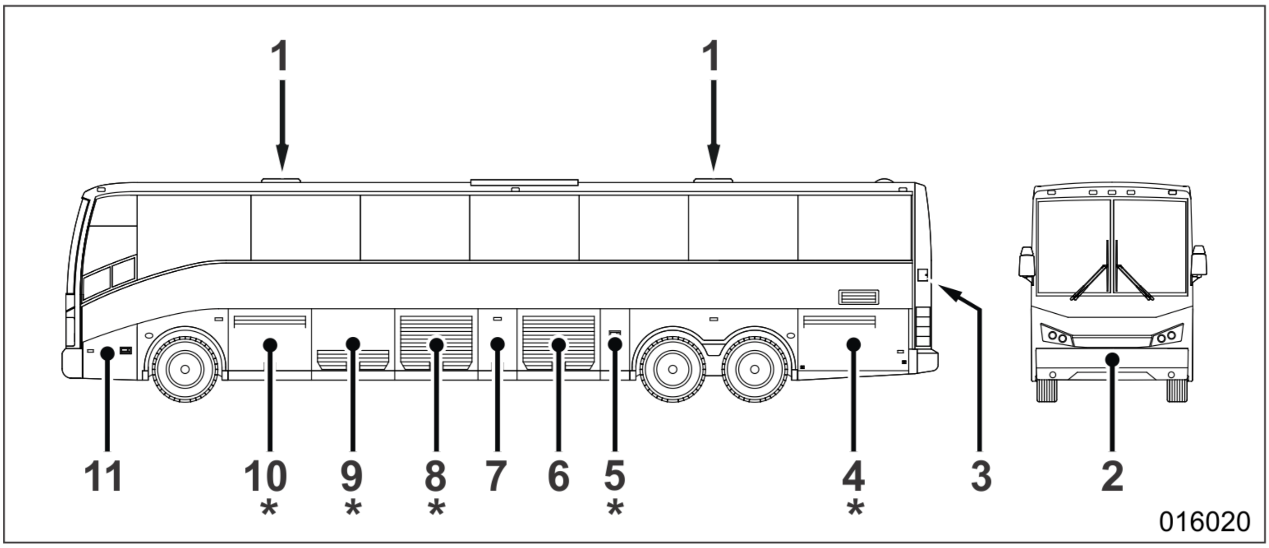
1. Identification / recognition

1.1 Access doors and controls at the outside *(continued)*

Left-hand side and front view

WARNING!

Compartments with an asterisk (*) contain high-voltage components.

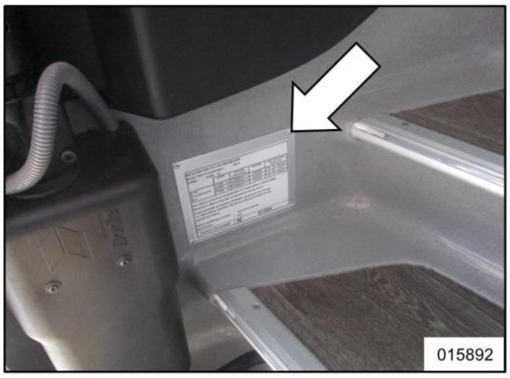


1. Roof hatch	**7.** Coolant-expansion tank of electro-components cooling circuit (traction system)
2. Spare wheel, compressed-air system fill fitting, air suspension test fittings of axle	**8.** Cooling group (radiators and cooling fans) of electro-components cooling circuit and brake resistors cooling circuit/interior heating circuit
3. Access to coolant-expansion tank filler cap of traction batteries cooling circuit	**9.** Brake resistors, cooling group of traction batteries, coolant-expansion tank of brake resistors cooling circuit/interior heating circuit, main shut-off valve in interior heating supply line, main shut-off valve in interior heating return line, interior-heating valve block
4. Traction batteries (two strings of three)	**10.** Voltage inverters of accessories, traction batteries (one string of three)
5. Refrigerant compressor	**11.** Windshield washer tank, front-bumper release lever
6. Refrigerant condenser compartment	

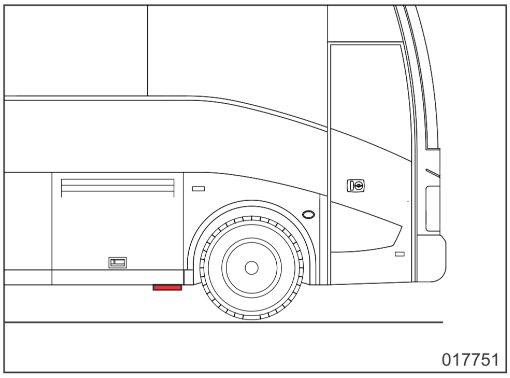
1. Identification / recognition *(continued)*

1.2 Vehicle identification number (VIN)

The vehicle identification number (VIN) can be found on a plate which is located at the left in the front entrance. It has also been stamped on the chassis side member behind the RH front wheel and is accessible from underneath.

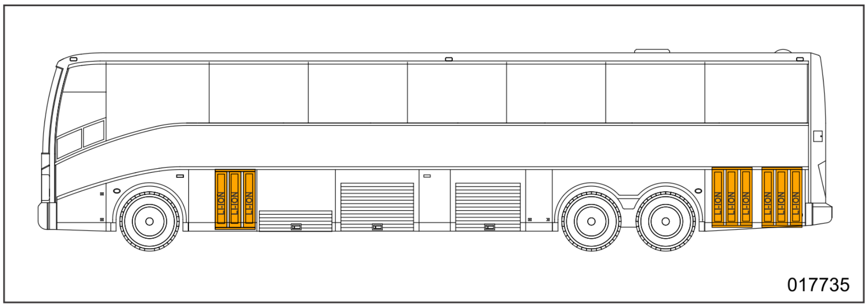


Vehicle identification plate in front entrance



Location of vehicle identification number on chassis side member

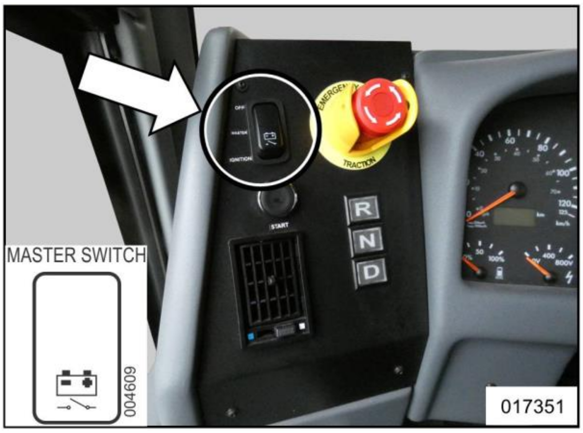
1.3 Traction battery identification

Make	Proterra
Chemistry family	Lithium-ion
Nominal voltage	650 VDC
Location of traction batteries	

LHS view of frontmost traction batteries (string of three)

RHS view of frontmost traction batteries (string of three)

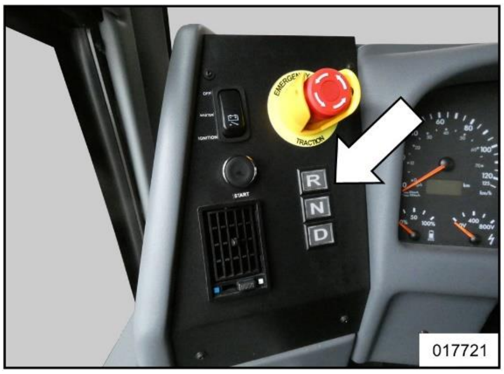
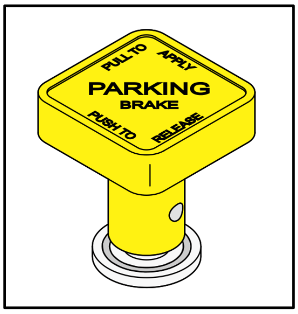
2. Immobilization / stabilization / jacking points



MASTER SWITCH positions	
OFF	24V electrical system off
MASTER	24V electrical system on
IGNITION	24V electrical system and ignition on

“MASTER SWITCH” on LH dashboard console

2.1 To immobilize vehicle

Step	Action
1	Put the traction system into neutral (N) by means of the travel-direction selector. 
2	Apply the parking brake by pulling the yellow parking brake knob at the LHS of the driver’s seat. 
3	Put the master switch in the “MASTER” position and wait at least eight seconds.
4	Set the master switch in the “OFF” position.

2.2 To stabilize vehicle

If the slope of the surface on which the tires are placed in relation to the road is greater than 30°, additional measures must be taken to support the vehicle against overturning.

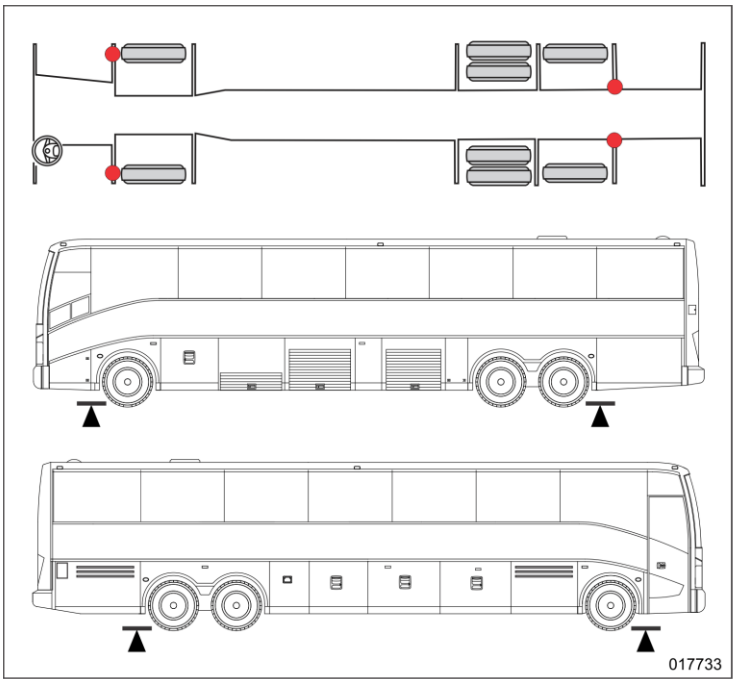
2. Immobilization / stabilization / jacking points *(continued)*

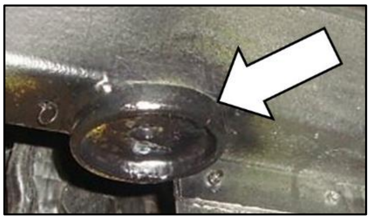
2.3 To lift vehicle

WARNING!

Never support the vehicle with hydraulic jacks only; always place chassis stands or support blocks, before working under a jacked up vehicle.

If the vehicle has to be raised by means of jacks under the chassis frame, position the jacking ram under the points indicated in the figure only. Always jack simultaneously at both front or both rear jacking points. The jacking points are indicated at the outside with triangle adhesives.





Detail of jacking point

If the front axle needs to be raised only, then place the jack inside the recess in the steering knuckle carrier, as indicated.

To raise an axle end of the front suspension, place jack here

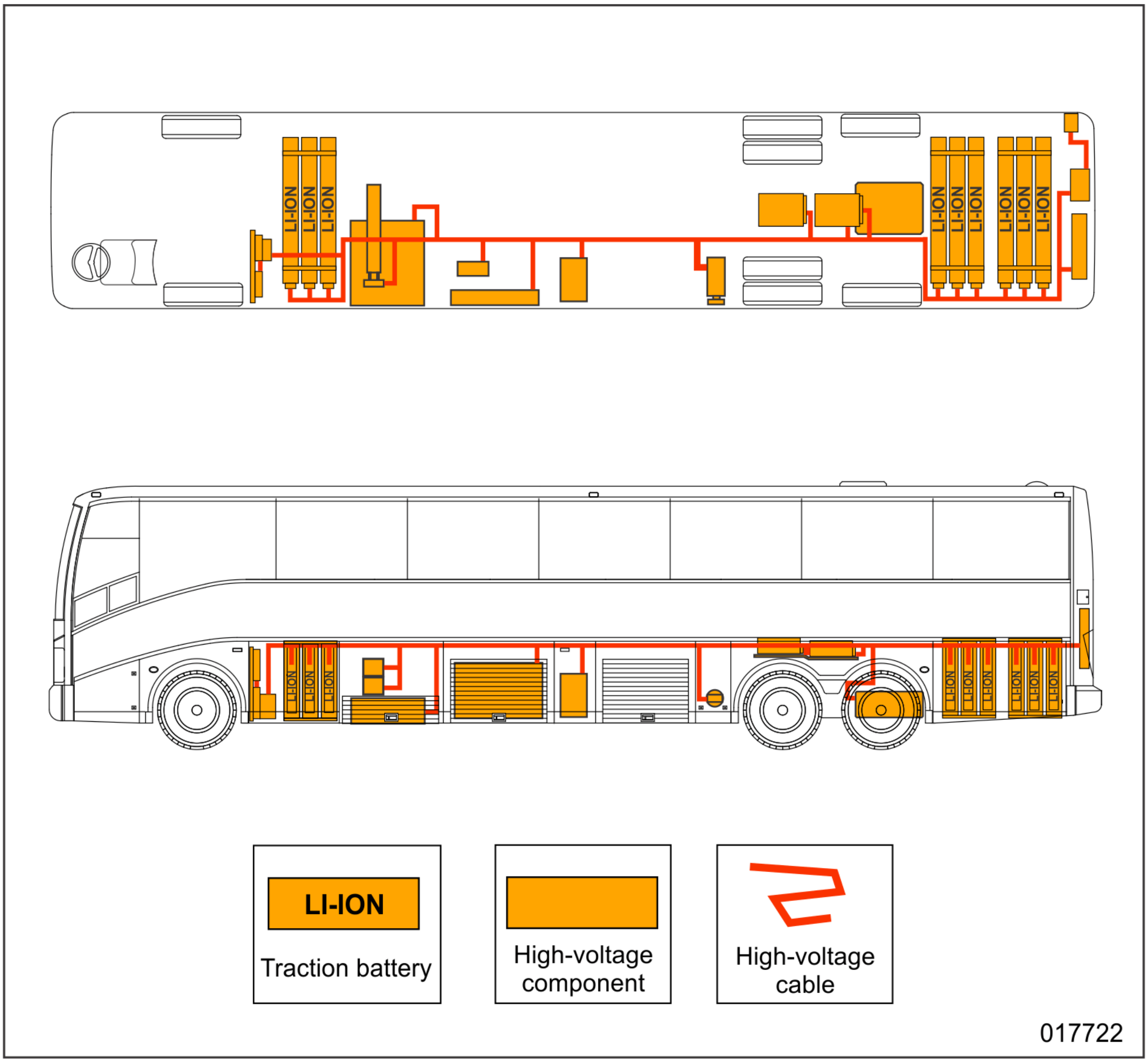
If the drive axle needs to be raised only, it is best to use a central jack with a dished head under the middle of the axle banjo.

3. Disable direct hazards / safety instructions

3.1 High voltage

Because of the presence of high voltage, danger of electrocution should always be reckoned with, as high-voltage components or cables can be damaged after an incident. Always de-energize the high voltage system first before any other measure has been taken. Always take into account the necessary personal protection equipment (PPE). Do not touch electrical components before the high-voltage circuits have been de-energized.

3.1.1 Location of high-voltage components/cables



NOTE: High-voltage cables are colored red in upper illustration to make a distinction with the other high-voltage components.

3. Disable direct hazards /safety instructions *(continued)*

3.1.2 Safety instructions regarding high-voltage components

As long as the traction system is switched on and up to 5 minutes after it has been switched off, a dangerously high voltage (maximum 750V) can be present at the following components of the system:

- traction batteries (DC);
- traction motor (AC)
- voltage inverters of traction system (DC);
- brake resistors (DC);
- junction box EK20 (DC);
- Voltage inverters of 400VAC and 24VDC electrical system (AC and DC);
- electric motor of air compressor (AC);
- electric motor of steering pump (AC);
- electric motor of interior cooling refrigerant compressor (AC);
- electric motor of coolant circulating pump (AC);
- cooling fans of traction system (DC).

NOTE: High-voltage cables are identifiable by the orange color.

WARNING!

Components under high direct voltage can cause an arc if touched or in case of spark-over. Danger of severe burns and electrocution. The arc will only extinguish by increasing the arc gap or by removing the voltage.

WARNING!

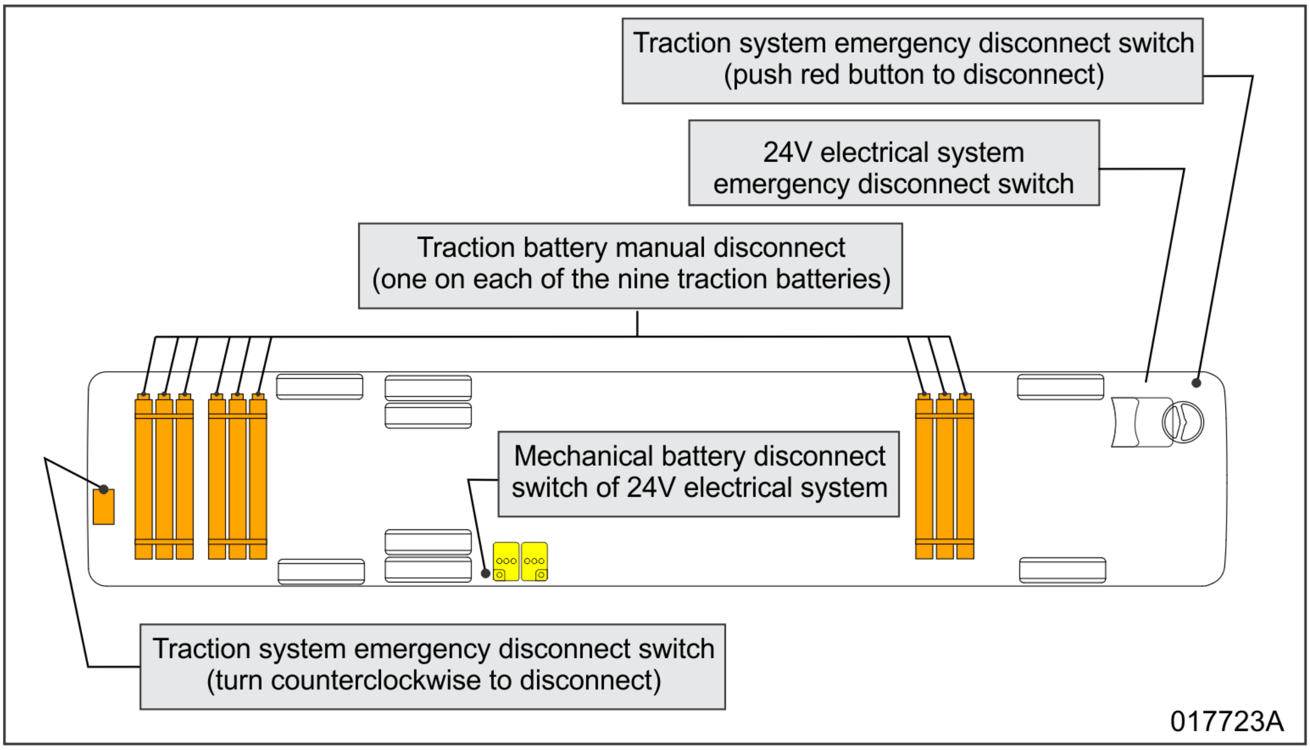
The traction batteries may contain a potentially lethal electrical charge that cannot be switched off. It is forbidden to open a traction battery.

WARNING!

After switching off the traction system, the capacitors of the voltage inverters can still contain a life-threatening electrical charge. As a rule, respect a discharge time of at least 5 minutes before starting to work on high-voltage components.

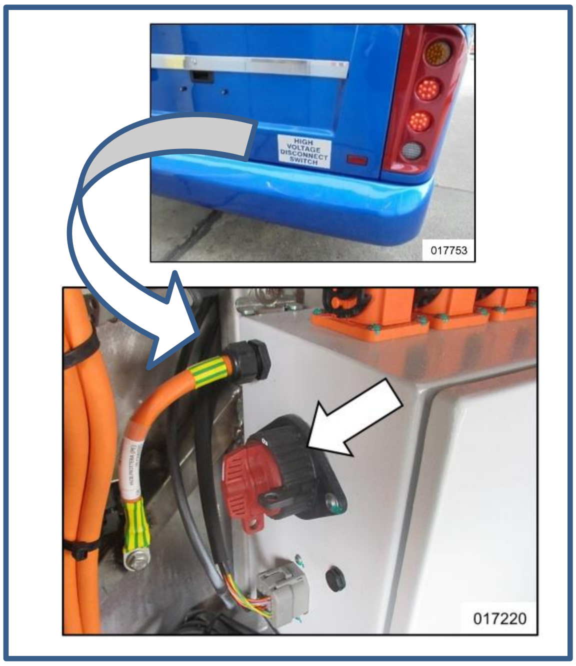
3. Disable direct hazards /safety instructions *(continued)*

3.1.3 To de-energize high-voltage circuit in case of emergency



Step	Action
1	Operate the emergency disconnect switch in the driver's cab ***(preferred)*** or the emergency disconnect switch behind the access door in the rear wall of the vehicle ***(alternative)***.

Emergency disconnect switch in driver's cab

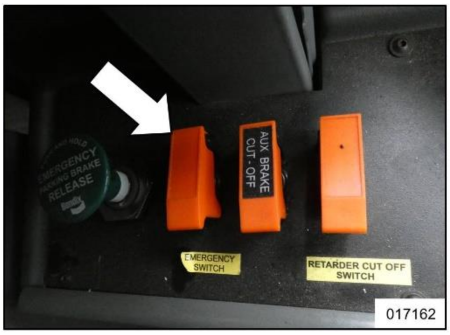


Emergency disconnect switch behind access door in rear wall of vehicle (indicated with a decal at the outside)

3. Disable direct hazards /safety instructions *(continued)*

3.1.3 To de-energize high-voltage circuit in case of emergency *(continued)*

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>2</td><td>If applicable: disconnect external voltage sources which have been connected to the vehicle (charging device for traction batteries, …).


<table>
<tr><th colspan="2">How to disconnect traction batteries charge cable?</th></tr>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1</td><td>Press the button on the handle of the charging cable to unlock it.</td></tr>
<tr><td>2</td><td>Pull the charging cable out of the socket on the vehicle.</td></tr>
</table>
Traction batteries charging control panel (behind the access door at the RHS (or LHS) at the very rear</td></tr>
<tr><td>3</td><td>Switch off the 24V electrical system by turning over the 24V emergency disconnect switch on the dashboard (preferred) or by turning the handle of the 24V mechanical battery isolation switch a quarter of a turn counterclockwise (alternative).


24V emergency disconnect switch on dashboard (at the LHS of the driver's seat)


24V mechanical battery isolation switch (in the compartment just ahead of the right-hand wheel of axle 2)</td></tr>
</table>

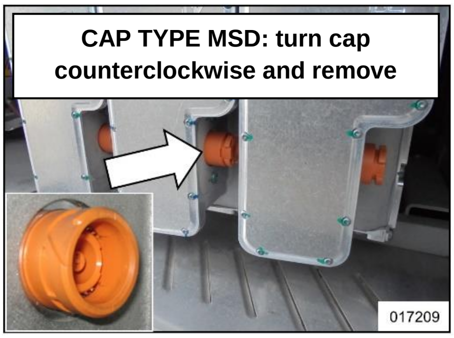
3. Disable direct hazards /safety instructions *(continued)*

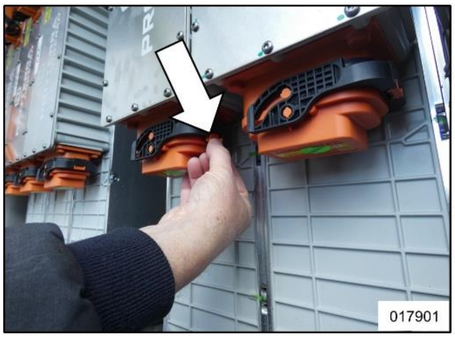
3.1.3 To de-energize high-voltage circuit in case of emergency *(continued)*

4

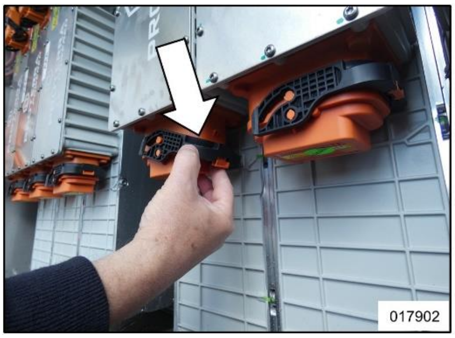
Wear “Class 0 HV” insulating gloves!

It is recommended to remove on each of the nine traction batteries the orange manual service disconnect (MSD). Two types of MSD are possible. The traction batteries are now mechanically isolated from the traction system.

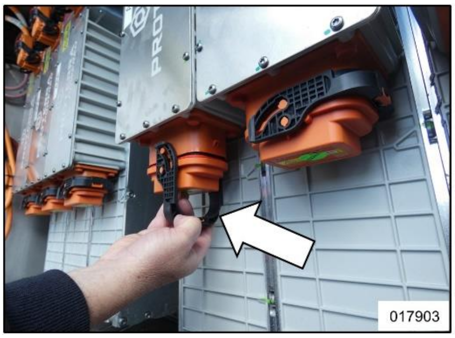




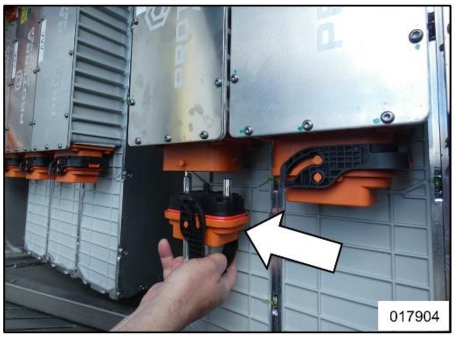
1. Push the small locking tab to unlock the lever of the MSD.



2. Turn the lever of the MSD down.



3. Turn further until stop.



4. Remove the MSD.

5

WARNING!

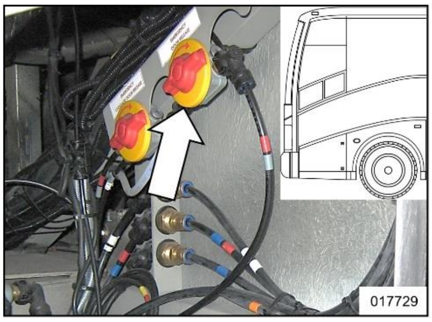
Wait at least 5 minutes. This is the time necessary to discharge the capacitors in the voltage inverters of the traction system.

4. Access to occupants

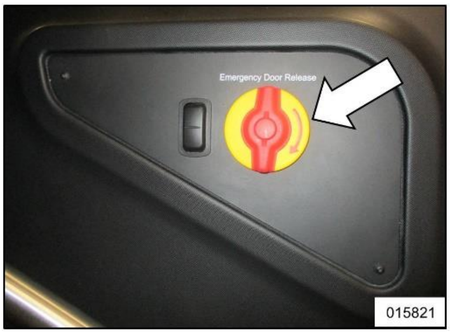
4.1 Passenger door emergency release

Emergency handles have been installed on the inside and outside of the vehicle in the vicinity of the passenger door. With these handles, it is possible to depressurize the passenger door actuating cylinder.

4.1.1 To open front passenger door in case of emergency



Exterior emergency handle of front passenger door (behind exterior access door below driver's window)

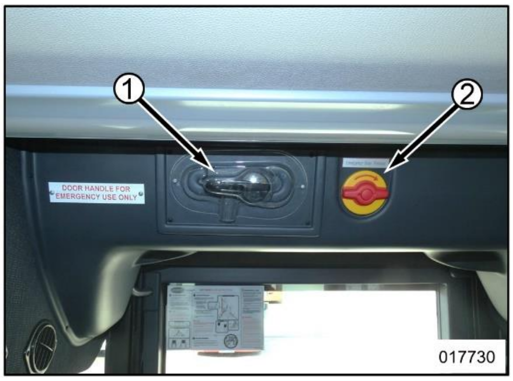


Interior emergency handle of front passenger door (at the RHS in entrance)

Step	Action
1	Turn handle a quarter of a turn clockwise to depressurize the door actuating cylinder.
2	Wait until the passenger door has lowered.
3	Push or pull the passenger door open.

4.2 Lift door emergency release (only at inside of vehicle)

An emergency handle has been installed at the inside above the lift door. With this handle, the pneumatic locking device can be depressurized.



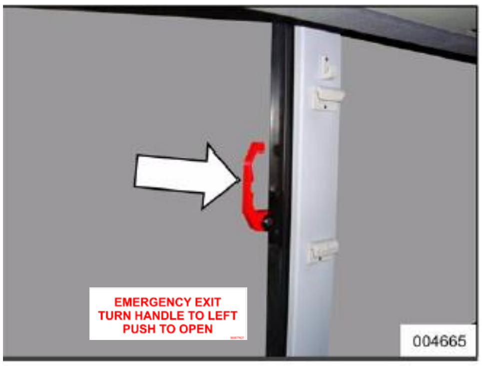
At the inside of the vehicle, above the lift door
1. **Emergency lever to manually unlatch lift door from inside**
2. **Emergency handle to depressurize pneumatic locking system of lift door**

Step	Action
1	Turn emergency handle (2) a quarter of a turn clockwise to depressurize the pneumatic locking system.
2	Break the plastic cover to get access to lever (1).
3	Position the lever on the lift door lock and unlatch the door.

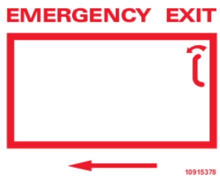
4. Access to occupants *(continued)*

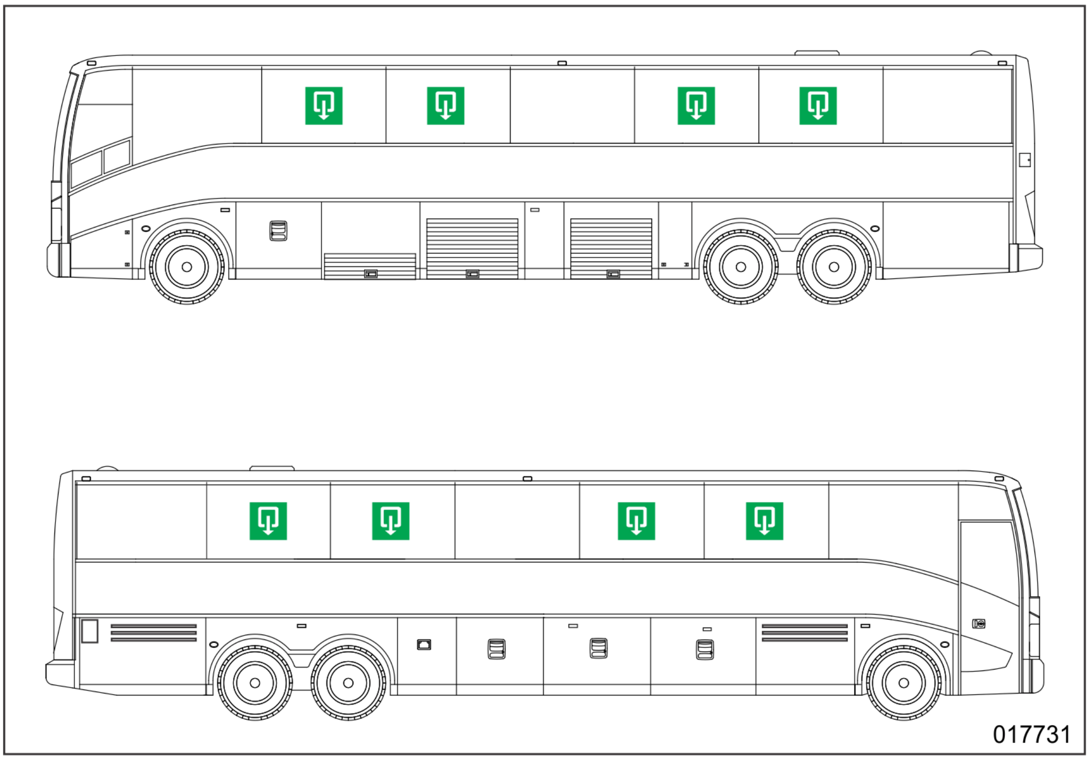
4.3 Emergency escape through side windows

A number of side windows can be used as emergency escape. The emergency side-windows are hinged at the top and have been provided with a handle in order to unlatch them from inside in case of an emergency. The emergency side-windows are marked with a red decal "Emergency exit". Simply pull the handle down and push the window outward.



Non-emergency side windows have decals indicating the nearest emergency exit.



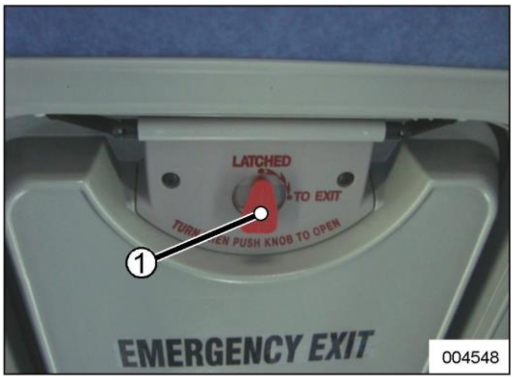


4. Access to occupants *(continued)*

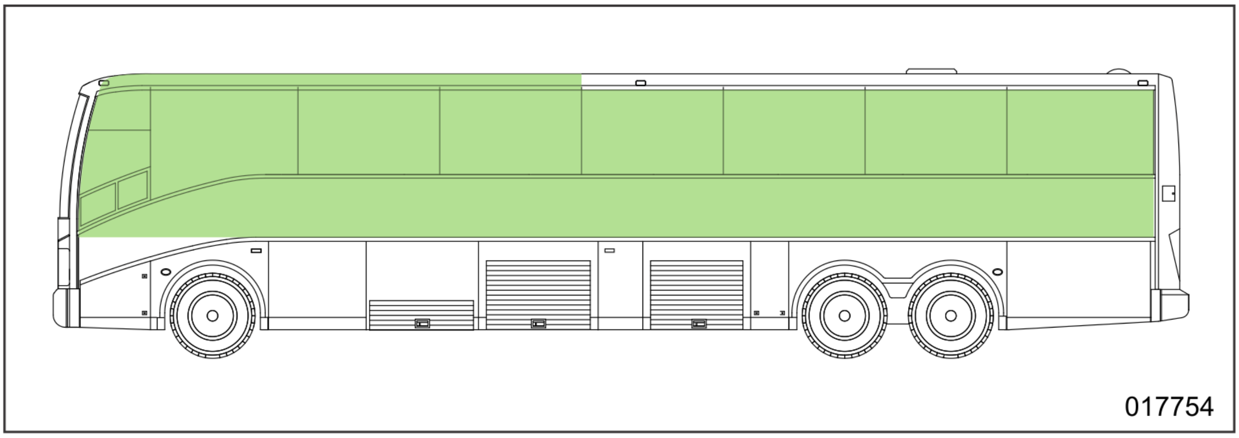
4.4 Emergency escape through roof hatch

The vehicle is equipped with two roof hatches which can be opened from the inside as to be used as emergency escape.

Step	Action
1	Push out hatch completely
2	Turn emergency knob (1) to position “TO EXIT” and push the knob outward.

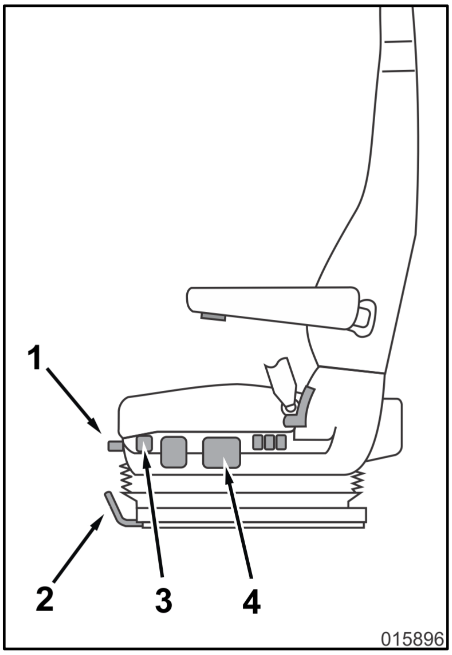


4.5 Metal structure cut zone



4.6 Position adjustment of driver’s seat

Function	Control	Action
To slide the seat cushion to the front or to the rear.	1	Pull the handle upward and slide the seat cushion.
To slide the complete seat to the front or back.	2	Pull the handle upward and slide the seat.
To move seat to lowest height.	3	Pull the handle down.
To change seat height.	4	Higher: pull the handle upward. Lower: pull the handle down.

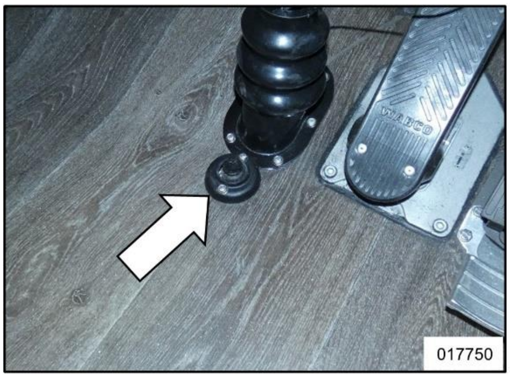


4. Access to occupants *(continued)*

4.7 Position adjustment of steering wheel

The steering wheel is adjustable in height as well as in tilt.

To change the steering wheel position:

Step	Action
1	Press and hold the unlocking knob in the footwell. 
2	Re-position the steering wheel.
3	Release the unlocking knob to lock the steering wheel again.

4.8 Safety belt cutters

The driver’s seat and all passengers seats are equipped with safety belts. Safety belt buckles unlock easily by simple pressing the button. In the event the buckle does not release a belt cutter can be used (two are present in the vehicle, one behind the driver’s seat and one at the rear of the vehicle).

4.9 Restroom

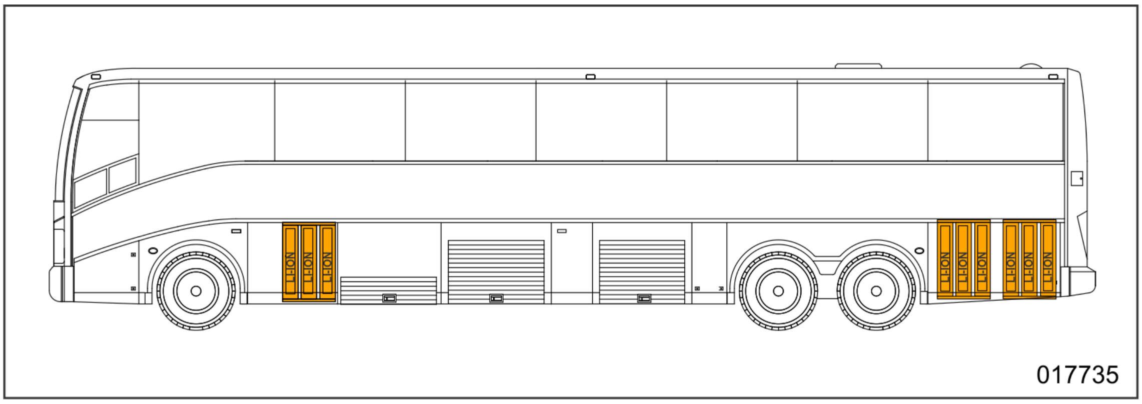
The vehicle is equipped with a restroom which is lockable from the inside. Next to the restroom entrance, a key of the door is provided so that in case of an emergency the door can be unlatched from the outside.

5. Stored energy / liquids / gasses / solids

5.1 Traction batteries

Make	Proterra
Chemistry family	Lithium-ion
Nominal voltage	650 VDC
Dangers	

Location of traction batteries



5.1.1 General first aid measures & environmental aspects

Under normal operational conditions, and provided that the traction battery pack enclosure remains closed, the traction batteries do not pose an electrical hazard and are provided with numerous safeguards to ensure that the high-voltage battery is kept safe and secure. Likewise the vehicle has been constructed in accordance to SAE J2910 and the general principles of ECE R100. Within the high-voltage system only touch safe connectors are used, high-voltage cables are orange and several markings indicating the presence of high voltage (in accordance with ISO 3864-1 and ECE R100) are present.

5.1.1.1 Exposure to high voltage (voltage higher than 60 V)

Every exposure to high voltage is life threatening, and even when the system is turned and de-energized it is likely to contain a substantial electrical charge that can cause injury or death when mishandled. This risk is further present when a traction battery is damaged or its enclosure is compromised then appropriate practice of high voltage preventive measures should be taken till the danger has been assessed and dissipated if necessary.
Exposed person to high voltage electric shock, take following first aid measures

- Do not approach the person till the power has shut down (see section 3)
- Check the condition of the person, is the person unresponsive then
- Check the airways and open if needed
- Check its breathing conditions
- Call 911 and perform CPR if needed.

5. Stored energy / liquids / gasses / solids *(continued)*

5.1.1 General first aid measures & environmental aspects *(continued)*

5.1.1.2 Exposure to material / electrolyte mixture leakage

WARNING!

Avoid contact with electrolyte.

Electrolyte, upon contact, will be irritating to the eyes and skin. Avoid contact until a positive identification can be made and sufficient protective equipment can be obtained (eye, skin and respiratory protection).

Use chemical strip to identify the spilled liquid; electrolyte will contain petroleum and organic solvent and fluoride compounds.

In the case that contact with skin has been made, danger for skin irritation and chemical burns is present. Flush immediately the contacted area of skin and wash appropriately the area with soap and water. If the irritation or chemical burns occurs or persist, seek immediately medical assistance.

In the case that contact has been made with the eyes then flush immediately with significant amounts of water for 15 minutes without rubbing and seek medical care by a physician at once.

5.1.1.3 Exposure to Material / Electrolyte mixture venting

When the batteries are exposed to abnormal heating or other abuse conditions, electrolyte and other electrolyte decomposition products can vaporize and be vented from cells. Such venting are to be considered as being an indicator of a thermal runaway reaction which is in itself an abnormal and hazardous condition.

When gases or smoke are observed escaping from the battery pack, evacuate the area and notify a first responder team. These vented gases are likely to be highly flammable and therefore could ignite unexpectedly as the condition that caused the venting could easily be the ignition source also. Approach to such a venting battery should be done with extreme caution and only by trained first responders equipped with proper personal protective equipment.

Vented gases may contain:

- Volatile organic compounds (e.g. alkyl carbonates, methane, ethylene, and ethane)
- Hydrogen gas
- Carbon dioxide
- Carbon monoxide
- Soot
- Particulates containing oxides of nickel, aluminum, lithium, copper, cobalt.
- Danger for forming vapors of Phosphorus pentafluoride, POF3 and HF

5. Stored energy / liquids / gasses / solids *(continued)*

5.1.1.3 Exposure to Material / Electrolyte mixture venting *(continued)*

WARNING!

Avoid contact with vented gases.

Vented gases may irritate eyes, skin, and throat. These cell vented gases are typically hot (may reach temperatures approximately of 600°C (1100°F), and may cause thermal burns.

In the case the vented gases and electrolyte vapors are inhaled, the person concerned should be placed and supplied with fresh air. If the person is not breathing perform artificial respiration and seek immediate medical assistance. Always consult a physician.

5.1.1.4 Always contact medical assistance

Upon contact with electrolytes or inhalation, always seek medical assistance and treatment.

5.1.1.5 Safety measures (for handling)

In the event of a damaged vehicle or any suspicion of leakage, always use proper protective gear and always shut down the vehicle as described under section 3. Every person should be aware that even when the high voltage system is turned off, the risk for exposure to high voltage remains.

The battery pack must never be intentionally short circuited, punctured, incinerated, crushed, immersed, force discharged or exposed to temperature above the declared temperature range of the product. An internal or external short circuit can cause significant overheating and provide an ignition source resulting in fire, including surrounding materials or materials within the cell or battery.

WARNING!

The battery assembly cover should never be breached or removed under any circumstances, including fire. Doing so might result in severe electrical burns, shocks, or electrocution.

5. Stored energy / liquids / gasses / solids *(continued)*

5.2 Coolant

Make	TEXACO Havoline XLC
Type	Ethylene glycol
Dangers	See (M)SDS attached 
Quantity (three separate cooling circuits) • Expansion tank of traction batteries cooling circuit • Expansion tank of traction system electro-components cooling circuit • Expansion tank of brake resistors cooling system and interior heating circuit	 approx. 14 liter (3.7 gal) approx. 5 liter (1.3 gal) approx. 14 liter (3.7 gal)

5.3 Refrigerant circuit in cooling group of traction batteries cooling circuit

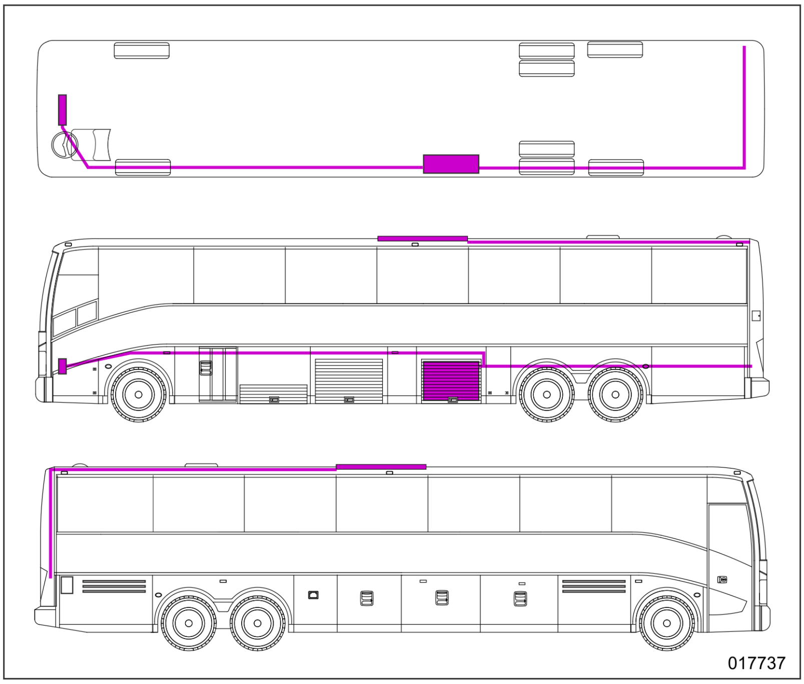
Refrigerant	R134a
Quantity	1.1 kg (2.4 lbs)
Dangers	See (M)SDS attached 

5.4 Drive axle

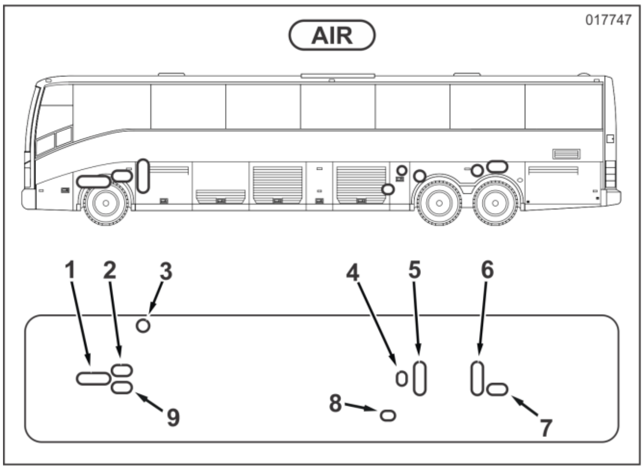
Type	Total transmission DUAL 9 FE
Quantity	approx. 17 liter (4.5 gallons)

5. Stored energy / liquids / gasses / solids *(continued)*

5.5 Refrigerant circuit of climate control system

Refrigerant	R134a
Quantity	12 kg (27 lbs)
Dangers	See (M)SDS attached 
Location of R134a containing components	

5.6 Compressed air

Number of compressed-air tanks	9
Maximum pressure	approx. 9 bar (130 psi)
Dangers	
Location of tanks **1.** Front axle brakes 29 liter (7.6 gal) **2.** Parking brake emergency release 20 liter (5.3 gal) **3.** Accessories 29 liter (7.6 gal) **4.** Buffer tank 4.72 liter (1.25 gal) **5.** Drive axle service brakes 29 liter (7.6 gal) **6.** Rear raising 29 liter (7.6 gal) **7.** trailing axle brakes 20 liter (5.3 gal) **8.** Air drying regeneration 4.72 liter (1.25 gal) **9.** Kneeling system 20 liter (5.3 gal)	

5. Stored energy / liquids / gasses / solids *(continued)*

5.7 Low voltage batteries (24V electrical system)

Electrolyte	Sulphiric acid
Quantity	Two 12V batteries, connected in series (= 24V)
Dangers	See (M)SDS attached 

5.8 Windshield washer cleaning fluid

Type	Water with windshield-washer anti-freeze
Quantity of tank	approx. 20 liter (5.3 gallons)

5.9 Power steering

Type	TOTAL Fluide ATX
Quantity	approx. 9 liter (2.5 gallons)

5.10 Restroom

Toilet type	Water flush

6. In case of fire

6.1 Automatic fire suppression system

The vehicle is not equipped with a fire suppression system.

6.2 Fire detection system

The vehicle is equipped with a fire detection system. The detection lines of the system have been located above traction batteries, steering pump electric motor and traction system voltage inverters. The system will, upon detection of a possible fire, inform the driver.

6.3 Which extinguishing method to use?

Small fire	Use ABC dry chemical extinguisher 	
Full fire and traction battery fire		Mandatory use of water to extinguish the fire
		ABC dry chemical extinguishers should only be used in case of the absence of water, or in the time before water is available.

6.4 What to do in case of a full fire or traction battery fire?

Always use the safety measures as explained under sections 2 and 3 to ensure a more safely adequate battling of the fire.

- **Maintain a safe perimeter (at least 20 feet) around the vehicle**

  As a traction battery implies high temperatures, toxic gases and a risk of explosions, a safety perimeter of at least 20 feet should be applied. As such responders should be aware that a fire within the traction battery goes from cell to cell which will lead to combustible electrolytes and graphite being blown off, together with periodic explosive flames as a result of the vaporizing, combustible electrolyte. Such explosive flames will present a serious risk for ejection of metal parts. It should be noted that the traction batteries are easy reachable by fire men allowing better control of the fire incident.

6. In case of fire *(continued)*

6.4 What to do in case of a full fire or traction battery fire? *(continued)*

- **Use an abundance of water to control the fire**

  Water allows the traction battery to cool and slow down the thermal runaway. But it can only be managed if the traction battery is continuously sprayed with water. In the event no water is immediately available, other agents like foam or CO can be used till water is available.
  Fire men are recommended to use an abundance of water (if possible) directly on the traction battery. If a reachable opening of the traction battery is present (e.g. vents or fractures from a collision) then these are to be used to apply water to the inside of the battery. This will force the battery to reduce its energy more rapidly and thus lowering the risk of re-ignition. But never open the battery with the intent to cool it!

- **Be aware that a traction battery may vent dangerous and toxic gases through controlled vents**

  As the traction battery will be subjected to rapidly increasing temperatures, toxic fumes (caused by combustible electrolytes and graphites) will appear and the risk of inflammation of these materials will increase. Such toxic fumes may consist of volatile organic compounds (VOCs), hydrogen gases, carbon dioxide, soot, and particulates containing oxides of nickel, aluminum, lithium, copper and cobalt. Additionally phosphorus pentafluoride, POF3 and HF vapors may form. The traction batteries have predefined vents to slow down any thermal runaway process through ejection of heat but also possible toxic fumes. The vehicle is designed to have an efficient airflow within the traction battery compartments so that no high concentration of these fumes remain within the vehicle when it is in motion.
  On the vehicle, ventilation shafts are present on each side and underneath the vehicle together with protection against rainfall and dirt. These shafts are to be used to spray water in and eventually opening the compartments

- **Use adequate personal protection gear**

  If a traction battery is ruptured or damaged in such a way that venting of gases and or electrolyte is possible or present, the use of a self-contained respiratory mask is highly recommended. Such apparatus are known as SCBA. The use of safety (neoprene or normal rubber) gloves and safety goggles is mandatory when handling damaged battery packs. Fire protective turnout gear should be worn as a lithium-ion battery fire can reach very high temperatures (600°C (1000°F)) when not controlled

6. In case of fire *(continued)*

6.4 What to do in case of a full fire or traction battery fire? *(continued)*

- **Make use of thermal imaging to check that all fire or heat sources are extinguished**

As traction battery fires can take up to 24 hours to extinguish completely, the use of thermal imaging equipment is highly recommended. Such equipment will allow to actively search and measure the temperature of the traction battery, and monitor its heating or cooling down evolution. In order for fire men to leave the hot zone, the traction battery should not be emitting any fire, smoke, or heat for at least one hour.

A quick summary of possible signs:

- Smoke still coming out of the traction battery;
- Noise like hissing, whistling, or rattling;
- Flying sparks and explosive flames coming from the traction battery;
- Abnormal aromatic odor.

Only then secondary responders (law enforcement, vehicle transportation,…) can safely enter the hot zone.

Due to the nature of the cause of compromising a traction battery of the vehicle (e.g. crash, submersion, fire,..) the excess water can be drained out by the second responders by tilting or repositioning the vehicle. The traction batteries should be monitored throughout this process as the danger for re-ignition remains present.

WARNING!

Always be aware that even after using the safety procedures a traction battery will always contains a high risk for electrocution.

- **Risk of self re-ignition**

As explained above cells can have a thermal runaway and the usage of thermal imaging should be used. Before secondary responders can enter the hot zone no sign of fire, smoke or heat should be present for at least one hour.

The usage of abundance of water is therefore the means to cool down a traction battery and slowing down the process. If the battery is cooled and kept cool then the thermal process will have stopped.

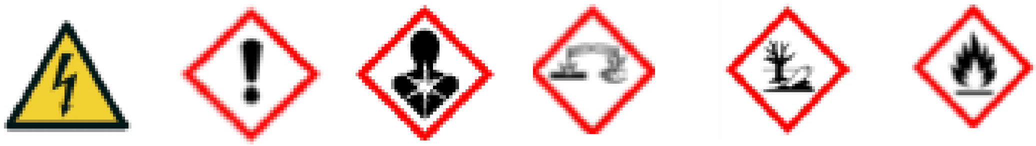
Always be aware of the risk of self-re-ignition.

7. In case of submersion

If the vehicle would submerge due to an accident or crash please be aware of the possibility that a high voltage is still present.

- Always use PPE and start with following the procedure as described within section 3.
- Beware that damaged batteries may self-ignite if a thermal process would start.
- Use thermal imaging to check the temperature evolution as prescribed under section 6.
- Always treat the vehicle (once out of the water) and its batteries as damaged till proper assessment has been performed.

High Voltage Lithium Ion batteries do not contain heavy metals such as lead cadmium or mercury. But the vehicle in itself and the batteries contain coolants and other liquids and materials which can be harmful to the environment.



8. Towing / transportation / storage

8.1 Towing instructions

8.1.1 Before towing

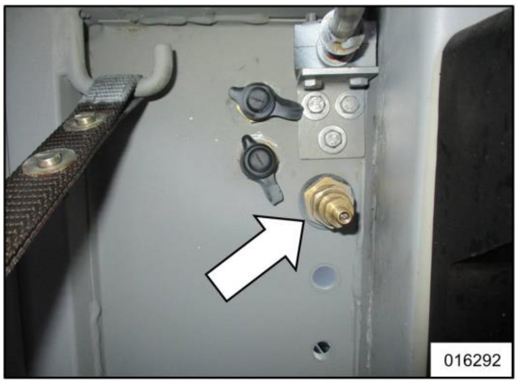
Before towing, always disconnect the propeller shaft at the drive axle end or remove the two drive-axle half shafts. In case of damage inside the drive axle, the vehicle has to be towed from the back.

8.1.2 To charge compressed-air tanks with an external supply

CAUTION!

Make sure the air is dry and clean. In addition, air pressure should not exceed 9 bar (130 psi).

During the towing procedure, the vehicle air system can be supplied with compressed air by the towing vehicle through a hose coupling. The coupling is located at the left behind the front bumper.



8.1.3 Parking brake emergency release

In case the vehicle is in need to be moved but has become immobile due to lack of pressure within the parking brake compressed air circuit which will automatically apply the parking brake. An emergency release knob is supplied to allow the vehicle to be moved as long this knob is pushed in.

CAUTION!

As this knob will activate supply of air through emergency release tank to the parking brake cylinders; its usage is limited to the amount of air provided within the tank.

8.1.4 Towing speed

CAUTION!

The towing speed must not exceed 60 km/h (37 mph).

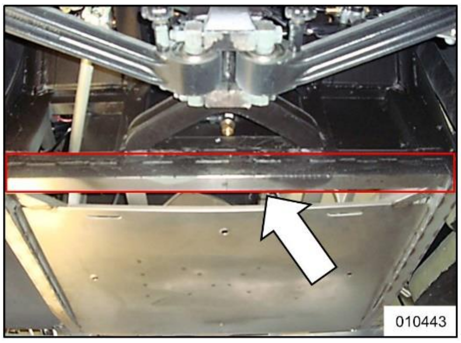
8. Towing / transportation / storage *(continued)*

8.1.5 To lift and tow vehicle with recovery vehicle equipped with under-arm lift

CAUTION!

Always take note of the towing instruction mentioned earlier.

Lift the vehicle at the front by placing the lift fork underneath the reinforced cross beam in front of the front axle.

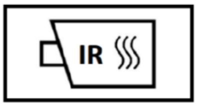


8.2 Storage

8.2.1 Storage of vehicle with traction batteries

If the vehicle has suffered from fire crash the possibility of damaged traction batteries exists and therefor a risk of re-ignition still exists, the vehicle must be stored in a secured area where it can be continiously be monitored or at a safe storage location when such life monitoring is not possible.

A safe storage location needs to be free of combustable materials and which is only accessible by trained professionals. Furthermore the temperature during storage. The area should be 50 feet downwind free of occupied structures. Maintain the use of thermal imaging to monitor the thermal condition of the vehicle on a regular basis.



Always contact service@proterra.com to obtain specific instructions before removing the traction batteries.

8.2.2 Storage of traction batteries

Never store damaged batteries next to undamaged batteries. Always handle batteries with extreme caution. Always follow the instructions given by service@proterra.com.

- Undamaged batteries
  Storage should be done under and maintained at ambient temperatures between -20°C and 30°C (-4°F and 86°F), humidity less than 70% and protected from condensation. The storage must have good ventilation, free from moisture and heat sources and well protected against flooding. Stored batteries should have a state of charge of 30% or less but not fully discharged for a period longer then 12 months. Storage areas need to be compliant with appropriate local fire codes.

- Damaged batteries
  During storage the damaged batteries must be monitored for signs of smoke, leakage of elektrolyte, leakage of coolant, or signs of heat. If full time monitoring is not possible, the batteries concerned should be moved to a safe storage location which will be free of flammable materials, accessible only by trained professionals and 50 feet downwind of occupied structures.

9. Important additional information

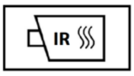
Always contact Proterra when the batteries are, or are suspected to be, damaged for specific instructions.

Always use appropriate PPE (in case of electrolyte leakage or suspicion of)
- Use safety glasses
- Use suitable protection clothing
- Use “Class 0 HV” insulation gloves
- Use self-contained full face respiratory mask
- Use insulated boots or use dielectric overboots

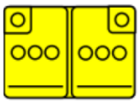
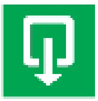
Contacts:

VANHOOL	ABC COMPANIES	PROTERRA
Bernard Van Hoolstraat 58 2500 Lier Belgium aftersales.bc@vanhool.com +32 03 420 20 20	Official distributor of Van Hool vehicles in the U.S. 17469 W. Colonial Drive Wintergarden, Florida 34787 1-877-427-7278	1815 Rollins Road Burlingame California 94010 service@proterra.com 1-800-255-3924

10. Explanation of pictograms used

	Mandatory use of water to extinguish the fire		Environmental hazard
	ABC fire extinguishers should only be used in case of the absence of water, or in the time before water is available		Harmful
	Use of thermal Infrared camera		Do not break open
	General warning sign		Safety perimeter
	Warning, electricity		Use Personal Protective Equipment (PPE)
	Warning low temperature		Steering wheel tilt control
	Explosive		Seat height adjustment
	Flammable		Seat adjustment, longitudinal
	Gases under pressure		Lifting point
	Corrosives		High-voltage cable
	Hazardous to human health		High-voltage component
	Acute toxicity		High-voltage traction battery

10. Explanation of pictograms used *(continued)*

	Low voltage battery (24V electrical system)
	Refrigerant line (air conditioning)
	Refrigerant component (air conditioning)
	Air tank
	Emergency window